

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (ME) Examination

November 2018

ME 401.01 FLUID MACHINES II

Date: 21.11.2018, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70.

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1** Answer the following objective questions. **[07]**

1. In centrifugal compressors, the ratio of outlet whirl velocity to the blade velocity is known as _____.
(a) work done factor (b) slip (c) degree of reaction (d) none
2. The usual value of a power input factor in centrifugal compressor is _____.
(a) 1.25 (b) 0.86 (c) 1.04 (d) 1.00
3. Which of the following type does Screw compressor belongs to?
(a) positive displacement compressor (b) dynamic compressors (c) both a & b (d) none of the above
4. The ratio of Isothermal power to actual measured input power of a compressor is known as _____.
(a) isothermal efficiency (b) volumetric efficiency (c) barometric efficiency (d) none
5. In case of three lobed roots of root blower air is sucked and compressed _____.
(a) two times (b) four times (c) six times (d) none of the above
6. Maximum speed of vane blower compressor may be _____ rpm.
(a) 12500 (b) 3000 (c) 2500 (d) 5000
7. The screw compressors are efficient in low air pressure requirements.
(a) agree (b) disagree

Q - 2 (a) Describe the working of scroll compressor with neat sketch. **[07]****(b)** Differentiate between reciprocating compressor and rotary compressor. **[04]****(c)** Explain rotating stall phenomenon in centrifugal compressor. **[03]****OR****Q - 2 (a)** Draw P-V diagram for the root blower and derive roots efficiency equation. Also list down the advantages. **[07]****(b)** Explain volute casing with neat sketch. **[03]****(c)** What do you mean by pre-whirl in centrifugal compressor. **[04]**

- Q - 3 (a)** Classify different types of compressors in details. [05]
- (b)** Determine the slip factor in a single sided centrifugal compressor fitted in the air craft flying with a speed of 230 m/s at an altitude where the pressure is 0.25 bar and the temperature 220 K. the mean diameter of the eye is 25.5 cm and the impeller tip diameter is 54 cm. Rotational speed of the compressor is 16000rpm and the inlet duct of the impeller eye contains fixed vanes which give the air pre-whirl of 25° at all radii. Stagnation pressure at the compressor outlet is 1.75 bar. Take the power input factor at 1.04 and isentropic efficiency as 0.8. [07]
- (c)** Define : Stagnation enthalpy and Stagnation temperature. [02]

OR

- Q - 3 (a)** Explain lift and drag of an Aero-foil. [05]
- (b)** Centrifugal compressor has a pressure ratio 4 with an isentropic efficiency of 90% when running at 11000 rpm and inducing air at 288 K. curved vanes at inlet gives the air a pre-whirl of 20° to the axial direction at all radial and the mean diameter of eye is 400mm. the absolute air velocity at inlet is 90m/s. The impeller tip diameter is 750mm. Calculate the slip factor. [07]
- (c)** Explain: Power input factor. [02]

SECTION II

- Q - 4** Answer the following objective questions. [07]
- The compressor capacity of a reciprocating compressor is directly proportional to its _____.
 (a) speed (b) pressure (c) volume (d) all
 - From the point of lower specific energy consumption, which of the following compressors are suitable for part load operation?
 (a) two stage reciprocating compressors (b) centrifugal compressors (c) two stage screw compressor (d) single stage screw compressor
 - Which of the following parameters are not required for evaluating volumetric efficiency of the compressor?
 (a) power (b) cylinder bore diameter (c) stroke length (d) FAD
 - The discharge temperature of two stage compressor as compared to single stage compressor is _____.
 (a) lesser (b) same (c) higher (d) none
 - A dynamic compressor works with a constant pressure.
 (a) true (b) false
 - An axial compressor gives optimum performance at high speeds and large volume flows.
 (a) true (b) false
 - Compressor will have higher efficiency if compression follows isothermal process.
 (a) true (b) false

- Q - 5 (a)** Calculate the diameter and stroke for a double acting single stage reciprocating air compressor of 50kW having induction pressure 100 KN/m² and temperature 150°C. The law of compression is $PV^{1.2} = C$ and delivery pressure is 500 KN/ m². The revolution/sec =1.5 and mean piston speed in 150 m/min. Clearance is neglected. **[07]**
- (b)** Explain the influence of intercooling on the cycle efficiency in multistage reciprocating air compressor? **[05]**
- (c)** Describe the steps required to perform a CFD study of a well-defined flow problem. **[02]**

OR

- Q - 5 (a)** A single acting reciprocating air compressor has cylinder diameter and stroke of 200 mm and 300 mm respectively. The compressor sucks air at 1 bar and 27°C and delivers at 8 bar while running at 100 r.p.m. Find: 1). Indicated power of the compressor; 2). Mass of air delivered by the compressor per minute; and 3). temperature of the delivered by the compressor. The compression follows the law $PV^{1.25} = C$. Take $R = 287 \text{ J/kg. K}$. **[07]**
- (b)** What is the effect of excessive clearance on the performance of an air compressor? **[05]**
- (c)** Which techniques are used for numerical discretization? **[02]**
- Q - 6 (a)** Write down different types of losses occurring in centrifugal fans. **[07]**

OR

- Q - 6 (a)** Enlist different types of centrifugal fans and explain any one with neat sketch. **[07]**
- (b)** Sketch and explain Aerofoil with detail nomenclature. **[05]**
- (c)** Define Magnus effect. **[02]**
- OR**
- (c)** What is CFD? Write down its different applications. **[02]**
